Julio Isidro y Sánchez

Quantitative Genetics · Genomic Selection · Genomic Mating · Plant Breeding

Tenured Scientist, Spanish National Research Council (CSIC)
Group Leader, Genomic-Assisted Breeding (The Rocinante Lab)
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Profile

Quantitative geneticist and plant breeder working on the statistical and computational methodology of genomic-assisted breeding. My research connects quantitative genetics with practical breeding decisions: mixed-model and genomic-prediction methodology, training-population and experimental-design optimisation, genotype × environment modelling, and genomic mating (optimal cross selection under constraints on genetic gain and diversity). The methods are released as open-source R packages (*MateR*, *TrainSel*, *GenomicMating*, *CovCombR*, *GEmetrics*) and are applied with public and commercial breeding programmes in cereals, sunflower, grapevine, berries and forest trees. Trained in crop physiology (PhD, Granada) and plant breeding (MSc, Cornell), with research positions in Canada, the USA, Ireland and Spain. Coordinator of the international Genomic Selection course since 2015.

Current position

2026–present	**Tenured Scientist (Científico Titular), Spanish National Research Council (CSIC)** *Centre for Plant Biotechnology and Genomics (CBGP, UPM–INIA/CSIC), Madrid* Group Leader, Genomic-Assisted Breeding group (The Rocinante Lab): three PhD students, three MSc students (CIHEAM Zaragoza) and one technician (October 2026).

Research expertise

Methodology	Quantitative genetics; mixed models and REML; genomic prediction (GBLUP, Bayesian and machine-learning models); multivariate and multi-environment models; selection indices and multi-objective optimisation.
Breeding design	Training-population optimisation; sparse testing and multi-environment trial design; two-stage analysis of historical data; genotype × environment interaction and stability.
Genomic mating	Optimal cross selection, usefulness criterion, gain–diversity trade-offs, optimal recycling of parents in breeding cycles.
Genetics	GWAS and QTL mapping; host–pathogen genomics (wheat–*Zymoseptoria tritici*, Fusarium, rusts); pan-genomics of oat and durum wheat.
Crops	Wheat (durum and bread), oat, barley, maize, sunflower, grapevine, blueberry, chickpea, sugarcane, lodgepole pine.

Research metrics

Publications	48 peer-reviewed journal articles, 5 book chapters; 54 documents indexed in Scopus (October 2026).
Citations	2,046 citations, *h*-index 20 (Scopus); 2,197 citations, *h*-index 21 (OpenAlex).
Funding	Principal investigator or co-PI on more than ten competitive projects (European, Irish, Spanish and industry funding) since 2016.
Training	6 PhD theses supervised (defended or in progress) and 9 MSc theses; 11 editions of the international Genomic Selection course since 2015.

Academic and professional positions

2026–present
Tenured Scientist, CSIC
Centre for Plant Biotechnology and Genomics (CBGP, UPM–INIA/CSIC), Madrid, Spain

2024–2026
Associate Professor (Profesor Titular de Universidad)
Universidad Politécnica de Madrid (UPM); CBGP

2020–2024
Beatriz Galindo Senior Distinguished Researcher
CBGP, Universidad Politécnica de Madrid
Competitive national programme of the Spanish Ministry of Science for the recruitment of researchers from abroad. Established the Genomic-Assisted Breeding group.

2015–2020
Lecturer / Assistant Professor in Crop Science
School of Agriculture and Food Science, University College Dublin (UCD), Ireland
Taught crop breeding, crop science and organic agriculture; led Irish and European projects on cereal genomics and genomic selection.

2015
Postdoctoral Researcher
Department of Plant Breeding and Genetics, Cornell University, USA
Genomic selection and training-population optimisation (with J.-L. Jannink and M. E. Sorrells).

2012–2015
MSc in Plant Breeding and Genetics (research interruption)
Cornell University, USA

2008–2012
Postdoctoral Fellow (NSERC Visiting Fellowship, 2011–2012)
Agriculture and Agri-Food Canada, Semiarid Prairie Agricultural Research Centre (SPARC), Swift Current, Saskatchewan
QTL mapping and quantitative genetics of agronomic traits in durum wheat (Canadian Wheat Improvement Flagship).

2007–2008
Research Technician
CSIC, Armilla (Granada), Spain

2003–2007
PhD Researcher (Ministry of Education and Science fellowship)
University of Granada, Spain

Education

2015
MSc in Plant Breeding and Genetics
Cornell University, Ithaca, USA

2008
PhD in Crop Physiology
University of Granada, Spain
Thesis: ecophysiological and molecular analysis of the impact of durum wheat breeding in Mediterranean environments on yield formation and grain protein and amino-acid accumulation (IDuWUE project).

2005
MSc in Agrobiology
University of Granada, Spain

2001
BSc in Biology
University of Seville, Spain

2017
Professional Teaching Certificate and Continuing Professional Development Certificate
University College Dublin, Ireland

Research funding

As principal investigator or coordinator

2025–2027
Combining genomic approaches to study host–pathogen relationships in wheat and Septoria [PI]
Spanish Research Agency, Consolidación Investigadora 2024, CNS2024-154812 · € 144K
Dual-genome prediction of *Zymoseptoria tritici* aggressiveness and wheat resistance; 200 wheat lines × 100 isolates; field validation.

2023–2026 **Climate-resilient blueberry varieties through genomic selection [PI]**
CDTI, IDI-20250085, collaboration with a horticultural breeding company · € 100K
Genomic selection for water-stress tolerance and warm climates over three breeding cycles (200–400 seedlings phenotyped per year).

2022–2025 **Genomic-assisted breeding for sustainable agriculture: a benchmark approach [PI]**
Spanish Research Agency, Generación de Conocimiento 2021, PID2021-123718OB-I00 · € 127K
Genomic selection benchmark with the AAFC Swift Current public wheat breeding programme (Canada).

2022–2026 **Machine-learning approaches applied to genomic-assisted breeding [PI]**
UPM Programa Propio predoctoral contract, PREDOC-21 · € 74K

2022–2026 **Genomic-assisted breeding applied to an industrial sunflower breeding programme [PI]**
Industrial Doctorate (Spanish Ministry of Science) with Syngenta · € 25K

2021–2024 **WheatRes: new sources of horizontal resistance to Septoria and rust in durum wheat [PI]**
Spanish Research Agency, Strategic Lines, PLEC2021-007930 · € 190K

2021–2023 **Breeding tools to harness yield productivity by applying genomic selection [PI]**
Comunidad de Madrid R&D programme · € 144K

2020–2023 **Healthy Oats: innovation in oat product development [Coordinator]**
Interreg Ireland–Wales; UCD, IBERS Aberystwyth, Swansea University, Teagasc · project coordinators J. Isidro y Sánchez and F. Doohan

2019–2022 **WheatSustain: knowledge-driven genomic predictions for sustainable disease resistance in wheat [PI (Ireland), WP1 lead]**
SusCrop ERA-NET Cofund; DAFM Ireland 2017EN104; 8 European institutions, coordinator M. Lillemo (NMBU) · € 1.5M total, UCD € 174K

2019–2023 **Developing multi-use barley to improve the organic Irish market [PI]**
Irish Research Council, Government of Ireland Postgraduate Scholarship GOIPG/2019/3601 · € 96K

2018–2020 **Effect of soil water content on seedling emergence in small-grain cereals [PI]**
European Plant Phenotyping Network (EPPN2020) transnational access

2016–2020 **Oats for the future: host resistance and RNAi to minimise mycotoxin contamination under present and future climates [co-PI]**
Science Foundation Ireland / BBSRC, 16/BBSC/3315; with F. Doohan, N. Magan and A. Medina · € 540K total, UCD share € 67K

2016–2021 **PICS: Physiology Infrastructure for Crop Stress [Co-investigator]**
Science Foundation Ireland Research Infrastructure, 16/RI/3747; coordinator A. Feechan · € 1.2M total, UCD € 295K

As team member or consortium partner

2019–2024 **InnoVar: next-generation variety testing for improved cropping on European farmland [Deputy WP2 leader]**
H2020 SFS-29-2018, grant 818144; 21 institutions · € 8M total, UPM € 319K
Tasks in WP2, WP3, WP4 and WP7 (genomics, phenotyping and machine learning for high-performance low-risk varieties).

2020–present **Oat PanGenome consortium [Member]**
International consortium; responsible for whole-genome sequencing of variety OT380 · € 36K

2020–present **Svevo Platinum Genome consortium [Member]**
International consortium improving the durum wheat reference genome · € 10K

2019–2023 **TRENDING_Wheat: accuracy and efficiency of selection for complex traits in wheat for Mediterranean environments [Team, GS implementation]**
Spanish Research Agency PID2019-109089RB-C32; PI J. M. Soriano (IRTA) · € 165K

2017–2022 **CONSUS: crop optimisation through sensing, understanding and visualisation [Funded investigator]**
Science Foundation Ireland; PI G. O'Hare · € 4M total; PhD studentship € 96K

2015–2019 **CTAG: Canadian Triticum Applied Genomics [Scientific advisor]**
Canadian consortium grant; design of a genomic selection programme · $10M total

2012–2017 **Improving wheat productivity under abiotic stress (Canadian Wheat Improvement Flagship) [Postdoctoral researcher]**
National Research Council Canada and AAFC · $14M total

2002–2009 **Durum wheat adaptation to Mediterranean environments [Team member]**
EU IDuWUE (ICA3-CT-2002-10085, € 963K); Spanish grants AGL2002-04285, AGL2005-07257, AGL2006-09226
Doctoral and early postdoctoral research on durum wheat physiology and QTL mapping.

Open-source software

R packages implementing published methods, developed in the group or with D. Akdemir; available on GitHub (TheRocinante-lab) and CRAN.

2025 **MateR**
*Genomic mating: usefulness criterion, optimal mate allocation and optimal recycling with control of genetic diversity [**fernandezgonzalez2026mater**]*
github.com/TheRocinante-lab/MateR

2024 **GEmetrics**
*Best linear unbiased prediction of genotype × environment metrics (superiority, stability) from genomic data [**fanellicarvalho2024stability**]*
CRAN; github.com/TheRocinante-lab/GEmetrics

2021 **TrainSel**
*Selection of training populations for genomic prediction, including sparse and multi-environment designs [**akdemir2021trainsel**]*
github.com/TheRocinante-lab/TrainSel

2020 **CovCombR**
*Combination of partially overlapping covariance and relationship matrices from heterogeneous data sources [**akdemir2020multiomics**, **akdemir2023em**]*
CRAN

2016–2018 **GenomicMating**
*Original implementation of genomic mating strategies [**akdemir2016mating**]; CRAN release 2018*
CRAN

2015 **STPGA**
*Selection of training populations by genetic algorithm [**akdemir2015ga**] (lead developer D. Akdemir)*
CRAN

Publications

48 peer-reviewed journal articles, 5 book chapters and 1 preprint. Own name in bold; consortium papers list the first authors and give the total number of authors. Years refer to the journal issue. Metrics: Scopus 2,046 citations, $h = 20$; OpenAlex 2,197 citations, $h = 21$ (October 2026).

Selected publications

2026 J. Fernández-González, S. M. Metwally and **J. Isidro y Sánchez**. MateR: a novel genomic mating framework. *Genetics* 232(4): iyag013. doi:10.1093/genetics/iyag013

2025 R. Avni, N. Kamal, L. Bitz, E. N. Jellen, W. A. Bekele and T. T. Angessa et al. A pangenome and pantranscriptome of hexaploid oat. *Nature* 649(8095): 131–139. Oat PanGenome consortium; 80 authors including J. Isidro y Sánchez. doi:10.1038/s41586-025-09676-7

2025 A. Domínguez-Rondón, R. Tirado, I. Solís, A. Sánchez-Vallet and **J. Isidro y Sánchez**. Integrating host–pathogen interaction genomics to boost wheat defense against septoria tritici blotch. *BMC Genomics* 26: 1066. doi:10.1186/s12864-025-12174-9

2025 J. Fernández-González and **J. Isidro y Sánchez**. Optimizing fully-efficient two-stage models for genomic selection using open-source software. *Plant Methods* 21: 9. doi:10.1186/s13007-024-01318-9

2024 A. Alemu, J. Åstrand, O. A. Montesinos-López, **J. Isidro y Sánchez**, J. Fernández-González, W. Tadesse, R. R. Vetukuri, A. S. Carlsson, A. Ceplitis, J. Crossa, R. Ortiz and A. Chawade. Genomic selection in plant breeding: key factors shaping two decades of progress. *Molecular Plant* 17(4): 552–578. doi:10.1016/j.molp.2024.03.007

2024 J. Fernández-González, B. Haquin, E. Combes, K. Bernard, A. Allard and **J. Isidro y Sánchez**. Maximizing efficiency in sunflower breeding through historical data optimization. *Plant Methods* 20: 42. doi:10.1186/s13007-024-01151-0

2023 J. Fernández-González, D. Akdemir and **J. Isidro y Sánchez**. A comparison of methods for training population optimization in genomic selection. *Theoretical and Applied Genetics* 136(3): 30. doi:10.1007/s00122-023-04265-6

2021 D. Akdemir, S. Rio and **J. Isidro y Sánchez**. TrainSel: an R package for selection of training populations. *Frontiers in Genetics* 12: 655287. doi:10.3389/fgene.2021.655287

2021 **J. Isidro y Sánchez** and D. Akdemir. Training set optimization for sparse phenotyping in genomic selection: a conceptual overview. *Frontiers in Plant Science* 12: 715910. doi:10.3389/fpls.2021.715910

2019 D. Akdemir, W. Beavis, R. Fritsche-Neto, A. K. Singh and **J. Isidro y Sánchez**. Multi-objective optimized genomic breeding strategies for sustainable food improvement. *Heredity* 122(5): 672–683. doi:10.1038/s41437-018-0147-1

2019 D. Akdemir and **J. Isidro y Sánchez**. Design of training populations for selective phenotyping in genomic prediction. *Scientific Reports* 9: 1446. doi:10.1038/s41598-018-38081-6

2016 D. Akdemir and **J. Isidro y Sánchez**. Efficient breeding by genomic mating. *Frontiers in Genetics* 7: 210. doi:10.3389/fgene.2016.00210

2015 D. Akdemir, **J. Isidro y Sánchez** and J.-L. Jannink. Optimization of genomic selection training populations with a genetic algorithm. *Genetics Selection Evolution* 47: 38. doi:10.1186/s12711-015-0116-6

2015 **J. Isidro y Sánchez**, J.-L. Jannink, D. Akdemir, J. Poland, N. Heslot and M. E. Sorrells. Training set optimization under population structure in genomic selection. *Theoretical and Applied Genetics* 128(1): 145–158. doi:10.1007/s00122-014-2418-4

Journal articles

2026 F. Afroze, N. Subramani, A. Kahla, C. McCabe, A. Feechan, A. Rahman, H. Fanelli Carvalho, **J. Isidro y Sánchez** and F. Doohan. Identification of germplasm and genomic markers associated with reduced T-2 and HT-2 mycotoxin accumulation in oat. *The Plant Genome* 19(4): e70321. doi:10.1002/tpg2.70321

2026 C. Copeland, **J. Isidro y Sánchez**, H. Fanelli and F. M. Doohan. Phenotypic and genetic resistance to Septoria blotch disease in European wheat varieties. *The Plant Genome* 19(2): e70237. doi:10.1002/tpg2.70237

2026 J. Fernández-González, J. Comadran Trabal, B. Haquin, A. Allard, E. Combes, K. Bernard and **J. Isidro y Sánchez**. Unveiling the interactions between design optimization, modeling strategies, and spatial analysis in genomic selection. *Theoretical and Applied Genetics* 139(9): 249. doi:10.1007/s00122-026-05285-8

2026 J. Fernández-González, S. M. Metwally and **J. Isidro y Sánchez**. MateR: a novel genomic mating framework. *Genetics* 232(4): iyag013. doi:10.1093/genetics/iyag013

2026 J. García-Abadillo, **J. Isidro y Sánchez**, D. Jarquín, E. Galeano and B. R. Thomas. Prediction of conelet mortality rates in interior lodgepole pine using phenomic, nutritional, and environmental data. *Frontiers in Plant Science* 17: 1845181. doi:10.3389/fpls.2026.1845181

2026 E. Mazzucotelli, C. Forestan, G. Zastrow-Hayes, D. Swarbreck, Y. Lev-Mirom and E. Cavalet-Giorsa et al. Durum wheat cv. Svevo reference genome Rel. 2.0: a comprehensive tool for wheat genomics. *Plant Biotechnology Journal* 24(10): 5460. Svevo Platinum Genome consortium; 73 authors including J. Isidro y Sánchez. doi:10.1111/pbi.70673

2026 J. Menor de Gaspar, R. D. Cuthbert, Y. Ruan, R. Knox, B. Fu, K. Wang, J. S. Sangha, S. Berraies, B. Meyer, F. E. Bokore and **J. Isidro y Sánchez**. Genetic control of wheat flour end-use quality and rheology by genome-wide association studies. *The Plant Genome* 19(2): e70236. doi:10.1002/tpg2.70236

2025 R. Avni, N. Kamal, L. Bitz, E. N. Jellen, W. A. Bekele and T. T. Angessa et al. A pangenome and pantranscriptome of hexaploid oat. *Nature* 649(8095): 131–139. Oat PanGenome consortium; 80 authors including J. Isidro y Sánchez. doi:10.1038/s41586-025-09676-7

2025 W. A. Bekele, R. Avni, C. L. Birkett, A. Itaya, C. P. Wight and J. Bellavance et al. Global genomic population structure of wild and cultivated oat reveals signatures of chromosome rearrangements. *Nature Communications* 16: 9486. 46 authors including J. Isidro y Sánchez. doi:10.1038/s41467-025-57895-3

2025 A. Domínguez-Rondón, R. Tirado, I. Solís, A. Sánchez-Vallet and **J. Isidro y Sánchez**. Integrating host–pathogen interaction genomics to boost wheat defense against septoria tritici blotch. *BMC Genomics* 26: 1066. doi:10.1186/s12864-025-12174-9

2025 J. Fernández-González and **J. Isidro y Sánchez**. Maximizing the accuracy of genetic variance estimation and using a novel generalized effective sample size to improve simulations. *Theoretical and Applied Genetics* 138(4): 78. doi:10.1007/s00122-025-04861-8

2025 J. Fernández-González and **J. Isidro y Sánchez**. Optimizing fully-efficient two-stage models for genomic selection using open-source software. *Plant Methods* 21: 9. doi:10.1186/s13007-024-01318-9

2025 J. Menor de Gaspar, A. Domínguez Rondón, J. García-Abadillo, R. Knox, F. E. Bokore, K. Boyle, K. Ammar, J. Huerta-Espino, S. Berraies, B. Meyer, W. Zhang, R. D. Cuthbert, F. Pierre, Y. Ruan and **J. Isidro y Sánchez**. Mapping novel yellow and leaf rust loci and predicting resistance in cross-derived Canadian durum wheat. *The Plant Genome* 18(4): e70124. doi:10.1002/tpg2.70124

2025 J. S. Sangha, W. Wang, R. Knox, Y. Ruan, R. D. Cuthbert, **J. Isidro y Sánchez**, L. Li, Y. He, R. DePauw, A. Singh, A. Cutler, H. Wang and G. Selvaraj. Phenotypic plasticity of bread wheat contributes to yield reliability under heat and drought stress. *PLOS ONE* 20(3): e0312122. doi:10.1371/journal.pone.0312122

2025 A. Viviani, J. K. Haile, W. G. D. Fernando, C. Ceoloni, L. Kuzmanović and D. Lhamo et al. Priority actions for Fusarium head blight resistance in durum wheat: insights from the Wheat Initiative. *The Plant Genome* 18(1): e20539. 36 authors including J. Isidro y Sánchez. doi:10.1002/tpg2.20539

2024 A. Alemu, J. Åstrand, O. A. Montesinos-López, **J. Isidro y Sánchez**, J. Fernández-González, W. Tadesse, R. R. Vetukuri, A. S. Carlsson, A. Ceplitis, J. Crossa, R. Ortiz and A. Chawade. Genomic selection in plant breeding: key factors shaping two decades of progress. *Molecular Plant* 17(4): 552–578. doi:10.1016/j.molp.2024.03.007

2024 A. Din, R. Gul, H. Khan, J. García-Abadillo Velasco, R. Persa, **J. Isidro y Sánchez** and D. Jarquín. Assessing the genotype-by-environment G×E interaction in desi chickpea via the Bayesian additive main effects and multiplicative

interaction model. *Agriculture* 14(2): 215. doi:10.3390/agriculture14020215

2024 H. Fanelli Carvalho, S. Rio, J. García-Abadillo and **J. Isidro y Sánchez**. Revisiting superiority and stability metrics of cultivar performances using genomic data: derivations of new estimators. *Plant Methods* 20: 85. doi:10.1186/s13007-024-01207-1

2024 J. Fernández-González, B. Haquin, E. Combes, K. Bernard, A. Allard and **J. Isidro y Sánchez**. Maximizing efficiency in sunflower breeding through historical data optimization. *Plant Methods* 20: 42. doi:10.1186/s13007-024-01151-0

2024 J. García-Abadillo, P. Adunola, F. S. Aguilar, J. H. Trujillo-Montenegro, J. J. Riascos, R. Persa, **J. Isidro y Sánchez** and D. Jarquín. Sparse testing designs for optimizing predictive ability in sugarcane populations. *Frontiers in Plant Science* 15: 1400000. doi:10.3389/fpls.2024.1400000

2024 J. García-Abadillo, P. Barba, T. Carvalho, V. Sosa-Zuñiga, R. Lozano, H. Fanelli Carvalho, M. García-Rojas, E. Salazar and **J. Isidro y Sánchez**. Dissecting the complex genetic basis of pre- and post-harvest traits in *Vitis vinifera* L. using genome-wide association studies. *Horticulture Research* 11(2): uhad283. doi:10.1093/hr/uhad283

2024 M. López-Fernández, A. Chozas, E. Benavente, E. Alonso-Rueda, **J. Isidro y Sánchez**, L. Pascual and P. Giraldo. Genome wide association mapping of end-use gluten properties in bread wheat landraces (*Triticum aestivum* L.). *Journal of Cereal Science* 118: 103956. doi:10.1016/j.jcs.2024.103956

2024 L. Morales, D. Akdemir, A.-L. Girard, A. Neumayer, V. K. Reddy Nannuru, F. Shahinnia, M. Stadlmeier, L. Hartl, J. Holzapfel, **J. Isidro y Sánchez**, H. Kempf, M. Lillemo, F. Löschenberger, S. Michel and H. Buerstmayr. Leveraging trait and QTL covariates to improve genomic prediction of resistance to Fusarium head blight in Central European winter wheat. *Frontiers in Plant Science* 15: 1454473. doi:10.3389/fpls.2024.1454473

2023 D. Akdemir, M. Somo and **J. Isidro y Sánchez**. An expectation-maximization algorithm for combining a sample of partially overlapping covariance matrices. *Axioms* 12(2): 161. doi:10.3390/axioms12020161

2023 J. Fernández-González, D. Akdemir and **J. Isidro y Sánchez**. A comparison of methods for training population optimization in genomic selection. *Theoretical and Applied Genetics* 136(3): 30. doi:10.1007/s00122-023-04265-6

2023 J. García-Abadillo, L. Morales, H. Buerstmayr, S. Michel, M. Lillemo, J. Holzapfel, L. Hartl, D. Akdemir, H. Fanelli Carvalho and **J. Isidro y Sánchez**. Alternative scoring methods of Fusarium head blight resistance for genomic assisted breeding. *Frontiers in Plant Science* 13: 1057914. doi:10.3389/fpls.2022.1057914

2022 S. Rio, D. Akdemir, T. Carvalho and **J. Isidro y Sánchez**. Assessment of genomic prediction reliability and optimization of experimental designs in multi-environment trials. *Theoretical and Applied Genetics* 135(2): 405–419. doi:10.1007/s00122-021-03972-2

2022 F. Shahinnia, M. Geyer, F. Schürmann, S. Rudolphi, J. Holzapfel, H. Kempf, M. Stadlmeier, F. Löschenberger, L. Morales, H. Buerstmayr, **J. Isidro y Sánchez**, D. Akdemir, V. Mohler, M. Lillemo and L. Hartl. Genome-wide association study and genomic prediction of resistance to stripe rust in current Central and Northern European winter wheat germplasm. *Theoretical and Applied Genetics* 135(10): 3583–3595. doi:10.1007/s00122-022-04202-z

2021 D. Akdemir, S. Rio and **J. Isidro y Sánchez**. TrainSel: an R package for selection of training populations. *Frontiers in Genetics* 12: 655287. doi:10.3389/fgene.2021.655287

2021 H. S. Hilmarsson, S. Rio and **J. Isidro y Sánchez**. Genotype by environment interaction analysis of agronomic spring barley traits in Iceland using AMMI, factorial regression model and linear mixed model. *Agronomy* 11(3): 499. doi:10.3390/agronomy11030499

2021 S. Rio, L. Gallego-Sánchez, G. Montilla-Bascón, F. J. Canales, **J. Isidro y Sánchez** and E. Prats. Genomic prediction and training set optimization in a structured Mediterranean oat population. *Theoretical and Applied Genetics* 134(11): 3595–3609. doi:10.1007/s00122-021-03916-w

2021 **J. Isidro y Sánchez** and D. Akdemir. Training set optimization for sparse phenotyping in genomic selection: a conceptual overview. *Frontiers in Plant Science* 12: 715910. doi:10.3389/fpls.2021.715910

2020 D. Akdemir, R. Knox and **J. Isidro y Sánchez**. Combining partially overlapping multi-omics data in databases using relationship matrices. *Frontiers in Plant Science* 11: 947. doi:10.3389/fpls.2020.00947

2020 **J. Isidro y Sánchez**, K. D'Arcy Cusack, C. Verheecke-Vaessen, A. Kahla, W. Bekele, F. Doohan, N. Magan and A. Medina. Genome-wide association mapping of *Fusarium langsethiae* infection and mycotoxin accumulation in oat (*Avena sativa* L.). *The Plant Genome* 13(2): e20023. doi:10.1002/tpg2.20023

2019 D. Akdemir, W. Beavis, R. Fritsche-Neto, A. K. Singh and **J. Isidro y Sánchez**. Multi-objective optimized genomic breeding strategies for sustainable food improvement. *Heredity* 122(5): 672–683. doi:10.1038/s41437-018-0147-1

2019 D. Akdemir and **J. Isidro y Sánchez**. Design of training populations for selective phenotyping in genomic prediction. *Scientific Reports* 9: 1446. doi:10.1038/s41598-018-38081-6

2018 S. Kumar, R. E. Knox, A. K. Singh, R. M. DePauw, H. L. Campbell, **J. Isidro y Sánchez**, F. R. Clarke, C. J. Pozniak, A. N'Daye, B. Meyer, A. Sharpe, Y. Ruan, R. D. Cuthbert, D. Somers and G. Fedak. High-density genetic mapping of a major QTL for resistance to multiple races of loose smut in a tetraploid wheat cross. *PLOS ONE* 13(2): e0192261. doi:10.1371/journal.pone.0192261

2017 D. Akdemir, J.-L. Jannink and **J. Isidro y Sánchez**. Locally epistatic models for genome-wide prediction and association by importance sampling. *Genetics Selection Evolution* 49: 74. doi:10.1186/s12711-017-0348-8

2017 **J. Isidro y Sánchez**, B. Perry, A. K. Singh, H. Wang, R. M. DePauw, C. J. Pozniak, B. L. Beres, E. N. Johnson and R. D. Cuthbert. Effects of seeding rate on durum crop production and physiological responses. *Agronomy Journal* 109(5): 1981–1990. doi:10.2134/agronj2016.09.0527

2016 D. Akdemir and **J. Isidro y Sánchez**. Efficient breeding by genomic mating. *Frontiers in Genetics* 7: 210.

doi:10.3389/fgene.2016.00210

2015 D. Akdemir, **J. Isidro y Sánchez** and J.-L. Jannink. Optimization of genomic selection training populations with a genetic algorithm. *Genetics Selection Evolution* 47: 38. doi:10.1186/s12711-015-0116-6

2015 **J. Isidro y Sánchez**, J.-L. Jannink, D. Akdemir, J. Poland, N. Heslot and M. E. Sorrells. Training set optimization under population structure in genomic selection. *Theoretical and Applied Genetics* 128(1): 145–158. doi:10.1007/s00122-014-2418-4

2012 **J. Isidro y Sánchez**, R. Knox, F. Clarke, A. Singh, R. DePauw, J. Clarke and D. Somers. Quantitative genetic analysis and mapping of leaf angle in durum wheat. *Planta* 236(6): 1713–1723. doi:10.1007/s00425-012-1728-5

2012 **J. Isidro y Sánchez**, R. Knox, A. Singh, F. Clarke, P. Krishna, R. DePauw, J. Clarke and D. Somers. Brassinosteroid leaf unrolling QTL mapping in durum wheat. *Planta* 236(1): 273–281. doi:10.1007/s00425-012-1603-4

2011 **J. Isidro y Sánchez**, F. Álvaro, C. Royo, D. Villegas, D. J. Miralles and L. F. García del Moral. Changes in duration of developmental phases of durum wheat caused by breeding in Spain and Italy during the 20th century and its impact on yield. *Annals of Botany* 107(8): 1355–1366. doi:10.1093/aob/mcr063

2008 F. Álvaro, **J. Isidro y Sánchez**, D. Villegas, L. F. García del Moral and C. Royo. Breeding effects on grain filling, biomass partitioning, and remobilization in Mediterranean durum wheat. *Agronomy Journal* 100(2): 361–370. doi:10.2134/agronj2007.0075

2008 F. Álvaro, **J. Isidro y Sánchez**, D. Villegas, L. F. García del Moral and C. Royo. Old and modern durum wheat varieties from Italy and Spain differ in main spike components. *Field Crops Research* 106(1): 86–93. doi:10.1016/j.fcr.2007.11.003

2007 C. Royo, F. Álvaro, V. Martos, A. Ramdani, **J. Isidro y Sánchez**, D. Villegas and L. F. García del Moral. Genetic changes in durum wheat yield components and associated traits in Italian and Spanish varieties during the 20th century. *Euphytica* 155(1–2): 259–270. doi:10.1007/s10681-006-9327-9

Book chapters

2022 **J. Isidro y Sánchez**, S. Rio and D. Akdemir. Hands-on training optimization in genomic selection. In: *Genomic Selection in Plants: A Guide for Breeders*. CRC Press. ISBN 9781032103501.

2020 **J. Isidro y Sánchez**, E. Prats, C. Howarth, T. Langdon and G. Montilla-Bascón. Genomic approaches for climate resilience breeding in oats. In: *Genomic Designing of Climate-Smart Cereal Crops*. Springer, Cham, pp. 133–169. doi:10.1007/978-3-319-93381-8_4

2017 G. Montilla-Bascón, C. D. Broeckling, O. A. Hoekenga, E. Prats, M. Sorrells and **J. Isidro y Sánchez**. Chromatographic methods to evaluate nutritional quality in oat. In: *Oat: Methods and Protocols*, Methods in Molecular Biology 1536. Springer, pp. 115–125. doi:10.1007/978-1-4939-6682-0_8

2017 **J. Isidro y Sánchez**, D. Akdemir and G. Montilla-Bascón. Genome-wide association analysis using R. In: *Oat: Methods and Protocols*, Methods in Molecular Biology 1536. Springer, pp. 189–207. doi:10.1007/978-1-4939-6682-0_14

2016 **J. Isidro y Sánchez**, D. Akdemir and J. Burke. Genomic selection. In: *The World Wheat Book, Volume III*. Lavoisier, Paris. ISBN 978-2-7430-2091-0.

Preprints

2025 S. M. Metwally, J. Fernández-González and **J. Isidro y Sánchez**. Expanding the gain–variance Pareto via optimal recycling and genomic mating. bioRxiv preprint. doi:10.1101/2025.09.28.679020

Teaching and training

International training

2015–present **Genomic Selection Course — coordinator and lead instructor**
gscourse.com; 11 editions; next edition May 2027, Universidad Politécnica de Madrid
One-week problem-based course for plant and animal breeders, graduate students and industry professionals: quantitative genetics, mixed models, genomic prediction, training-set optimisation, G×E, genomic mating; hands-on sessions in R with real breeding data. Co-instructors T. Olivoto (UFSM, Brazil) and J. Fernández-González (UPM). More than 500 participants from public and private breeding programmes to date.

2023–2024 **Evaluation of selection strategies**
CIHEAM Zaragoza (Mediterranean Agronomic Institute), Spain
Invited lecturer, MSc in Plant Breeding.

2017 **Bioinformatics to advance wheat breeding**
University of Bologna, Italy, 13–14 November 2017
Invited instructor.

University teaching

2020–present
Genomic-assisted breeding
MSc in Computational Biology, Universidad Politécnica de Madrid

2020–present
Molecular markers and their application in plant breeding
MSc in Agroforestry Biotechnology, Universidad Politécnica de Madrid

2020–present
Breeding for abiotic, biotic and quality traits
MSc in Biotechnology Applied to Plant Breeding, Universidad Politécnica de Madrid

2015–2020
University College Dublin, School of Agriculture and Food Science
Module coordinator and lecturer
Crop Breeding (CPSC30090); Introduction to Crop Science (CPSC10010); Fundamentals of Arable Crop Production (CPSC20090); Organic Agriculture (CPSC30050, 2016–2020); Emerging Crop Pathogens (CPSC30100, 2016–2018); Plant Physiological Ecology (CPSC20040, 2017–2018); Current Developments in Plant Biology (BOTN40200, 2017–2018).

2014–2015
Genetic Improvement of Crop Plants (PBLR4030)
Cornell University — teaching assistant

2005–2007
Crop Physiology laboratory
University of Granada — laboratory instructor

Supervision

Six PhD theses (defended or in progress) and nine MSc theses supervised; six postdoctoral researchers mentored in the last five years. Former group members now work at the University of Florida, University of Guelph, CIRAD, INRAE, Bayer and seed companies.

Doctoral theses

2024–2028
Juan Menor de Gaspar
Universidad Politécnica de Madrid
Genomic-assisted breeding applied to a public wheat breeding programme (with AAFC, Canada).

2024–2028
Alejandro Domínguez Rondón
Universidad Politécnica de Madrid
Data-driven plant breeding: computational methods and applications; host–pathogen genomics of wheat and *Zymoseptoria tritici*.

2022–2026
Javier Fernández-González
Universidad Politécnica de Madrid
Genomic-assisted breeding applied to an industrial sunflower breeding programme (Industrial Doctorate with Syngenta).

2022–2026
Julián García-Abadillo
Universidad Politécnica de Madrid
Machine-learning approaches applied to genomic-assisted breeding. Thesis

2019–2022
Eogan Curran
University College Dublin
Prediction of disease outbreaks based on wheat cultivar and pathogen genomic models.

2019–2022
Laura Paire
University College Dublin
Developing multi-use barley to improve the organic Irish market.

Master theses

2025–2026
Juan Manuel Gallego
UPM
Optimising genomic mate allocation for multi-trait wheat breeding.

2024–2025
Seifelden Metwally
CIHEAM Zaragoza
Optimising genomic mating strategies to enhance breeding outcomes (CIHEAM thesis award).

2023–2024
Alejandro Domínguez Rondón
UPM
Integrating genomics and genomic selection to enhance wheat resistance against *Septoria tritici*. Thesis

2023–2024 **Juan Menor de Gaspar**
UPM
Empirical data analysis for training population selection in wheat breeding. Thesis

2022–2023 **Bleck Tita Tabeh**
Universitat Politècnica de València
Optimisation of genomic selection models for Fusarium head blight resistance in wheat through nonlinear parameterisation. Thesis

2020–2021 **Javier Fernández-González**
UPM
Training optimisation in genomic selection: a comparison of algorithms. Thesis

2020–2021 **Pablo Atienza**
UPM
Copy-number variation in altitude-adapted *Zea mays* germplasm. Thesis

2017–2019 **Kane D'Arcy Cusack**
University College Dublin (MRes)
Association mapping of hexaploid oat cultivars for resistance to Fusarium mycotoxins.

2017–2019 **Simone Pasqualin**
University College Dublin (MRes)
Rate and timing of nitrogen application on agronomic and quality traits of *Eragrostis tef*.

Current MSc students (CIHEAM Zaragoza, 2026)

Habiba Youssef, Nour Selmi and Gökalp Karamaya.

Invited talks and seminars

2026 **International Symposium on Cereal Leaf Blights**, Canada, 26–29 May. Dual-genome prediction of wheat–*Zymoseptoria tritici* interactions enhances disease resistance breeding (speaker).

2025 **19th EUCARPIA Biometrics in Plant Breeding meeting**. A novel framework to control genetic diversity for optimal genomic mating (with S. Metwally and J. Fernández-González, speaker); Why REML variance estimates have no biological meaning and how to solve it (with J. Fernández-González, speaker).

2025 **Encuentros de Biología**, University of Málaga. Selección genómica: la revolución silenciosa en el mejoramiento de cultivos (invited lecture).

2022 **From Seed to Pasta IV**, Bologna, Italy, 26–29 October (invited speaker).

2022 **XVIII EUCARPIA Biometrics in Plant Breeding conference**, Gif-sur-Yvette, France, 21–23 September. Training population optimisation in genomic selection (invited speaker).

2022 **11th International Oat Conference**, Perth, Australia. Two invited talks: GWAS of *Fusarium langsethiae* infection and mycotoxin accumulation in oat; genomic prediction and training set optimisation in a structured Mediterranean oat population.

2020 **Monogram Network meeting**, James Hutton Institute, UK. Adventures in multi-omics: combining heterogeneous datasets via relationship matrices (invited).

2019 **IRTA Lleida** and **University of Galway** — invited seminars on genomic-assisted breeding and genomic selection for crop improvement.

2018 **Irish Cereals Improvement Network**. Cereal research and breeding for resistance: GWAS in wheat (invited).

2017 **Monogram Network meeting**, University of Bristol. Selecting training populations for genomic selection using STPGA (invited).

2016 **10th International Oat Conference**, 11–15 July. How should I select individuals of my training population for genomic selection? (invited).

2006 **University of Buenos Aires**, Seminarios de Cereales. Eco-physiological and molecular impact of durum wheat breeding on yield and grain composition (oral).

Conference contributions (selected)

	Posters and talks at PAG (2007, 2012, 2013, 2015, 2018, 2024, 2026), the International Conference on Quantitative Genetics (Vienna 2024, four posters from the group), International Wheat Conference (Perth 2024), EUCARPIA (2006, 2022, 2025), International Symposium on Grapevine Breeding and Genetics (2022), Canadian Wheat Symposia (2011, 2014), ITMI (2014), International Symposium on Pre-Harvest Sprouting (2011), Plant Genomics European Meetings (2006) and Spanish cereal and plant-breeding congresses (2004, 2007, 2019). Full list available on request.
2026	PAG 33, San Diego. Durum wheat cv. Svevo reference genome Rel. 2.0 (Mazzucotelli et al., poster).
2025	19th EUCARPIA Biometrics. Genetic control of wheat flour end-use quality and rheology via GWAS (Menor de Gaspar et al., poster).
2024	7th ICQG, Vienna. Integrating host–pathogen genomics to enhance wheat resistance against Septoria tritici blotch (Domínguez Rondón et al.); empirical insights into training set optimisation for AAFC's wheat breeding programme (Menor de Gaspar et al.); the role of sparse designs and advanced modelling in genomic selection (Fernández-González et al.) — posters.
2024	3rd International Wheat Conference, Perth. InnoVar outcomes: how genomics can characterise and increase the breeding value of durum varieties (Bozzoli et al., poster).
2024	PAG 31, San Diego. Svevo platinum genome and tetraploid wheat pangenome (Cattivelli et al.); global oat diversity (Bekele et al.); durum wheat gene-regulatory networks (Forestan et al.) — posters.
2022	XIII International Symposium on Grapevine Breeding and Genetics, Landau, Germany. GWAS in table grape breeding families (poster).
2018	PAG XXVI. Multi-objective optimised breeding strategies (Akdemir et al., poster).
2015	PAG XXIII. Optimisation of training populations under population structure in genomic selection (poster).

Industry collaboration and knowledge transfer

	Translation of quantitative-genetic methodology into operating breeding programmes. Confidential partner data are not disclosed; partners are named only where the collaboration is public.
2022–2026	**Syngenta** — sunflower breeding: industrial doctorate on historical-data optimisation, trial design and modelling strategies for genomic selection (three joint publications).
2023–2026	**Horticultural breeding company** — CDTI-funded implementation of genomic selection in a blueberry breeding programme.
2022–2026	**Agriculture and Agri-Food Canada**, Swift Current — genomic selection benchmark in a public durum and bread wheat programme.
2015–present	**Software transfer** — the group's R packages (MateR, TrainSel, GenomicMating, CovCombR, GEmetrics) are used by public and private breeding programmes in Europe, the Americas and Oceania; the Genomic Selection course serves as the main training channel.

International collaborations

Cornell University and Agriculture and Agri-Food Canada (training-set optimisation, wheat genomics); University of Florida (sparse testing, G×E modelling); BOKU Vienna and NMBU (Fusarium and rust prediction in European wheat); SLU Sweden, CIMMYT and ICARDA (genomic selection review); University of Bologna and the Wheat Initiative (durum wheat genomics, Fusarium head blight); Oat PanGenome and Svevo genome consortia; INRAE and CIRAD (former group members); University College Dublin, Teagasc, IBERS Aberystwyth and Swansea University (oat and wheat disease genomics); CIHEAM Zaragoza (MSc training).

Editorial and professional service

2018–2024	Associate Editor, *Spanish Journal of Agricultural Research* (section Plant Breeding, Genetics and Genetic Resources).
2015–present	Reviewer for *Nature*, *Theoretical and Applied Genetics*, *Heredity*, *Genetics Selection Evolution*, *The Plant Genome*, *Frontiers in Plant Science*, *Crop Science*, *Euphytica*, *Plant Breeding*, *Agronomy*, *Plant Communications*, *Journal of Agricultural, Biological and Environmental Statistics*, *Irish Journal of Agricultural and Food Research*.
2018–present	Vice-President, Spanish Research Society in Ireland (SRSI).
2020–present	Member (vocal), UPM research cluster *Agricultura Digitalizada y Sostenible* (Agri4DS).

Science outreach

2025	Encuentros de Biología, University of Málaga: public lecture on genomic selection in crop improvement.
2022	Cadena SER, *Hora 25*: "Semillas" (radio).
2019	RTVE, *Punto de Enlace*: genomic selection research in Ireland (radio); Native Scientist Dublin, science and language workshops for children; *Organic Matters* (Irish Organic Association): winter organic naked barley trial in Co. Wexford.
2018	Gestiona Radio, *Investigadores por el mundo*.

Fellowships and awards

2020	Beatriz Galindo Senior Distinguished Researcher fellowship, Spanish Ministry of Science (competitive national programme), CBGP–UPM.
2012	Talentia Fellowship, Regional Government of Andalusia.
2011	NSERC postdoctoral fellowship, Canada.
2009	Postdoctoral fellowship, Spanish Ministry of Education and Science.
2003	PhD fellowship, Spanish Ministry of Education and Science.
2025	Supervisor of CIHEAM Zaragoza MSc thesis award (S. Metwally).

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